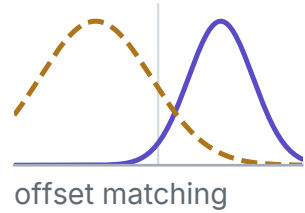


Coevolution, place by place



offset matching

coevolution

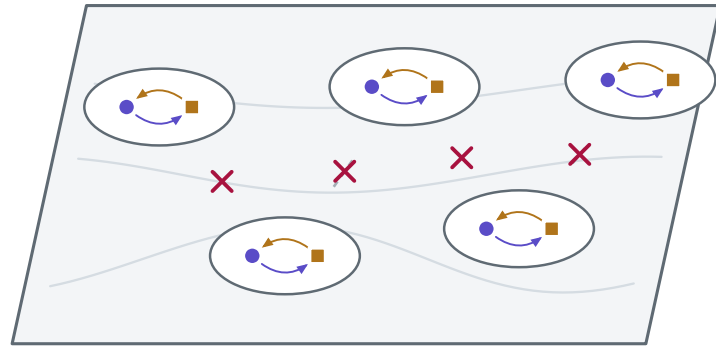
strengths B_1, B_2

abiotic selection

strengths A_1, A_2

drift

finite populations

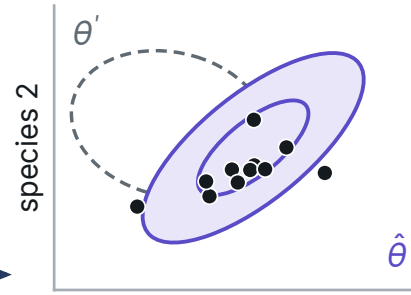


separate populations, no gene flow:

each one an independent run of the same process

The likelihood

predicted pattern against observed pattern



mean trait, species 1
one point per population

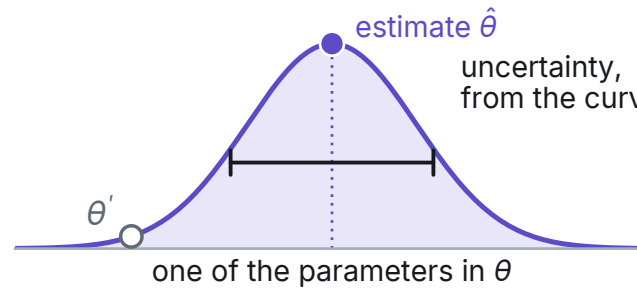
$$P(\text{data} \mid \theta) = \prod_i P(\bar{x}_i, \bar{y}_i \mid \theta)$$

**a product over populations,
because they are independent**

↓ read as
 $L(\theta \mid \text{data})$

the likelihood

θ : all the parameters,
 $B_1, B_2, A_1, A_2, \delta$



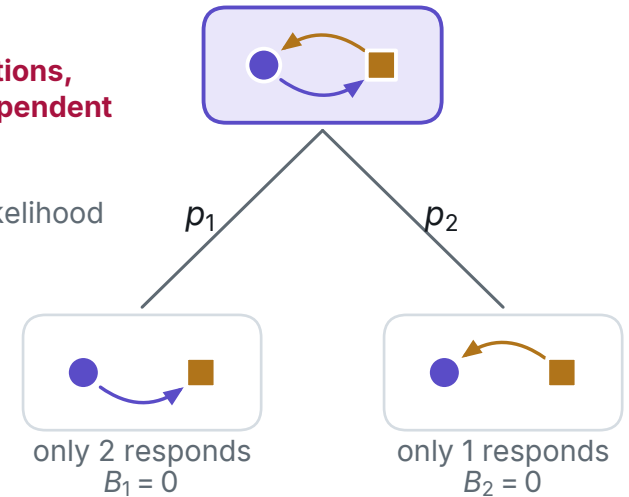
estimate $\hat{\theta}$

uncertainty,
from the curvature

one of the parameters in θ

A likelihood, and a test

coevolution: $B_1, B_2 \neq 0$



likelihood-ratio tests: coevolution is supported
only if both one-sided models are rejected